

Downstream Concentration in Regulated Vertical Relationships: Car Dealerships in Texas

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Abstract

This paper provides the first merger retrospective of automobile dealerships, using detailed transaction and registration data from Texas to study the competitive effects of increasing downstream concentration under state franchise law. Despite large changes in local market structure when dealerships are acquired by dealer groups, I find little to no impact on prices or quantities sold. Instead, the results uncover a previously undocumented mechanism through which manufacturers discipline dealer behavior despite statutory prohibitions on many standard vertical restraints. Manufacturers use discretionary vehicle allocations to shape dealers' product assortments, reallocating high-demand models and trims in ways that affect markups, sales, and investment incentives. Evidence from merger events, spillovers across dealerships, and a legal case study indicates that this mechanism mitigates downstream market power and alters the economic consequences of dealer consolidation. These findings challenge prevailing views of dealer consolidation and franchise regulation and imply that the manufacturer–dealer relationship may be more efficient than previously assumed.

Transactions of car dealerships are not typically large enough to trigger automatic reporting requirements as required by law. However, there is and has been active discussion within the industry of the effect of rising concentration begot by the rise of dealer groups.¹ Publicly traded companies such as Sonic, Group One, and Penske each own hundreds of car dealerships nationwide. Berkshire Hathaway counts the fifth-largest dealer group in the country as part of its holdings portfolio. One question that arises from this observation is whether regulators like state-level antitrust authorities should consider dealer merger enforcement as one of their regular enforcement activities.

¹See, e.g., “With Car Dealerships’ Ranks Thinned, the Survivors Thrive,” *The New York Times*, March 26, 2015; “Consolidation continues to reshape auto retail,” *Automotive News*, April 1, 2023; “Consolidation: Fewer Companies Operate Car Dealerships Every Year,” *Kelley Blue Book*, April 5, 2022.

The model of car distribution in the United States has long drawn scrutiny from academics and regulators. State laws enacted to ostensibly protect vulnerable dealers from the heft of their manufacturer counterparts have ultimately been maligned as political favoritism. Manufacturers face a near-complete ban on operating their own dealerships, and there are myriad ways in which incumbent dealers are shielded from competition or pressure from their franchiser. Smith 1982 in an early analysis shows that at a cross-sectional level, states that have more franchise restrictions tend to have higher car prices. Later, policy advocacy gained momentum as Tesla embarked on its crusade against the traditional franchise system. Lafontaine and Scott Morton 2010 offers a strong critique of the status quo of entrenched incumbent dealers and advocates principally for legalizing direct-to-consumer car sales; they also advocate for a loosening of franchise entry and exit restrictions that hamper manufacturers’ attempts to right-size intrabrand competition. The policy debate reached a head in 2014, when the Federal Trade Commission itself weighed in and published a blog post endorsing a direct-to-consumer model.² This drew a strong rebuke from the franchise dealers’ trade group, NADA, and after a years-long struggle played out in state legislatures, a status quo emerged which more or less maintained state-level restrictions but allowed Tesla to operate a viable distribution model.

Tesla has always and will always refuse to enter franchising relationships with dealerships and will only sell directly to consumers. They have won sparingly few concessions (notably in Virginia, as discussed by Scott Morton and McDermott 2017³). Yet in 2025, Tesla effectively operates a direct-to-consumer distribution system by maintaining only “showrooms” in most states and executing sales in states where they are allowed. For instance, in Texas, sales are executed in California and the customer registers the car in Texas (rather than the typical process where a dealer submits registration on behalf of the buyer). Other arrangements that have the semblance of direct-to-consumer sales, notably Hyundai’s partnership with Amazon to “sell” cars on the platform, ultimately delegate to an independent franchise dealer and hence fail in practice to disintermediate the dealer network.

The policy debate was reignited again in 2025 when a manufacturers’ trade association responded to a Department of Justice call for information on harmful state legislation with an open letter decrying state franchise laws.⁴ This drew another rebuke from NADA and marked the “holy war” between dealers and manufacturers as unsettled.⁵

²The blog post has since been removed, but coverage can still be found; see, e.g., Reuters, April 24, 2014: “FTC officials back Tesla’s direct-to-consumer car sales model.”

³See finding from Virginia DMV Commissioner Holcomb n.d. that there are “no independent dealers in the Richmond [...] area to own and operate a Tesla franchise in a manner consistent with the public interest.”

⁴See Alliance for Automotive Innovation: “RE: Request for Information on State Laws Having Significant Adverse Effects on the National Economy or Significant Adverse Effects on Interstate Commerce.”

⁵Automotive News, July 16, 2025: “Alliance, NADA clash in ‘holy war’ over state franchise laws in Justice Department review”

My focus on ownership consolidation among dealerships must inevitably contextualize this broader debate over the franchising model itself. Concerns about dealer groups' exercise of market power only arise because states have outlawed other retail models, and the concern is only so acute because of the limited threat of entry that incumbent dealers face.

In this paper, I use data on three years' worth of car dealership acquisitions in Texas to perform a merger retrospective and assess competitive effects from increasing concentration. Despite the large shifts in local market power generated when dealerships are transferred between dealer groups, I find that the transactions have little to no effect on prices or sales. Instead, my analysis of the economic and regulatory forces at work paints a nuanced picture of the franchising relationship between car manufacturers and their dealers. Despite the prohibition on many classical vertical restraints from the economic theory literature, manufacturers still appear to have a mechanism available with which to discipline the behavior of their dealers. This is the practice of allocations, in which manufacturers have some discretion to award their preferred dealer a better assortment of vehicles. I supplement my empirical evidence with a case study of this mechanism, which appears to violate the letter of the state franchise law but which, at least in Texas, has been deemed by courts as admissible conduct on the part of manufacturers. I also validate theoretical evidence from Lafontaine and Margaret E. Slade 1998 and suggestive evidence from Murry 2018 that the incentive contracting aspect of franchising is important here: When transacted dealers increase their sales, this has a spillover effect to other dealers of the same franchise. This implies some protection from intrabrand competition is necessary for the efficiency of the entire arrangement, and consolidation can increase total surplus because of better-aligned incentives. My conclusion is that despite the negative appearance of these state franchise laws and the trend of increasing ownership consolidation, the dealership system may be working closer to an "ideal world" than previously thought.

The paper is organized as follows: In the next section I discuss the legal history of state franchise laws, introduce my legal case study, and review the economic literature on franchising and vertical restraints. Then, I discuss the data sources I use in my analysis. I present the event study specification that underlies my analysis and present the results themselves. I include a discussion section regarding implications of these findings as well as a normative discussion on what a reasonable policy stance would be, taking these findings as given. A final section concludes.

1 Background

1.1 Legal Context

A particular path dependence in the legal history of car franchising has amounted to a patchwork of state laws that strongly favor dealers in today’s world. Franchising quickly became the dominant form of retail when the industry was new. Dealers were subject to the whims of larger and more dominant manufacturers, often abiding with sudden, unexpected franchise terminations and terms of trade generally bordering on extortionary (Lafontaine and Scott Morton 2010). This culminated with the Dealer’s Day in Court Act passed by the United States Congress in 1956, giving dealers legal recourse for unfair business practices on the part of the manufacturer. What these unfair practices *are* came to be outlined in state law, accumulating over time to a state-by-state chimera of regulations that tended to strongly favor dealers over manufacturers. The sum total of these restrictions has amounted to what Smith 1982 in an early analysis and Lafontaine and Scott Morton 2010 in a later commentary would call the entrenchment of car dealers, both in the classic market power sense and the political economic sense.

In Texas, Occupations Code §2301 regulates the relationship between dealers and manufacturers. Among the provisions laid out in this chapter are bans on quantity forcing, price discrimination, reduced threat of entry, reduced threat of franchise termination, and bans on direct manufacturer competition. In particular, Section 2301.451 prohibits quantity forcing, using the language that no manufacturer or distributor “may require or attempt to require” a dealer to order, accept delivery of, or pay for any motor vehicle that the dealer did not voluntarily order or contract for. Section 2301.468 forbids “inequitable treatment” of two franchised dealers “as a result of the application of a formula or other computation or process intended to gauge the performance of a dealership... .”

Given this legal framework, allocations would appear to be a prohibited practice. Allocation is the practice of portioning out the already-manufactured cars among the dealers in a manufacturer’s network. Allocations are typically determined at the level of regional management and consist of two parts—“formula” allocations and “discretionary” allocations. Formula allocations are supposed to account for latent or anticipated demand at a dealer’s location, which is a function primarily of the demographics of its sales territory. Discretionary allocations, as the name suggests, come at the discretion of regional management, and are typically rewards for meeting sales targets or for participating in manufacturer programs like facility renovation. These discretionary allocations appear to exactly be differential treatment of dealerships “as a result of the application of a formula or other computation or process intended to gauge the performance of a dealership... .” Despite the wealth of online discussion regarding the mechanics of how allocations work, there

is a comparative dearth of discussion regarding the legal justification for allocations. In practice, while a manufacturer controls allocations and sets sales standards, dealerships have recourse to challenge allocations via a formal protest and hearing before an administrative law judge (ALJ).

There is only one recent court proceeding in Texas that addresses the legality of discretionary allocations. In *Hyundai Motor America v New World Car Nissan Inc*, World Car Hyundai in San Antonio challenged their allocations between 2010 and 2013 as unfair. An ALJ ruled in favor of Hyundai, the DMV Board of Directors reversed the decision, and finally upon appeal, the Texas Court of Appeals in Austin in 2019 reversed the Board’s reversal and settled the matter for Hyundai. (The case was appealed to the Texas State Supreme Court, with no subsequent action on the court’s part.) The appellate court’s opinion hence gives the best insight available into where legal boundaries of control are drawn in this process.

The undisputed facts of the case are that World Car Hyundai did not participate in Hyundai-sponsored programs, with the one specifically mentioned being a program to renovate dealer facilities. Their competitor, Red McCombs Hyundai, did. These sponsored program specifically used discretionary allocations as an inducement to encourage dealers to participate. In a six-month period during this time, Red McCombs received 134 cars via discretionary allocation, whereas World Car received 20. In addition, Hyundai employed a sales-efficiency standard to determine additional discretionary allocations, one which World Car Hyundai could not meet with its inventory levels. The conclusion to which the ALJ and appellate court came is that it “was reasonable for Hyundai to reward dealers that participated in Hyundai-sponsored programs and renovated their facilities with extra discretionary allocation.” Regarding sales efficiency, the ALJ and appellate court found that it was reasonable because the sales target was not a *requirement* for World Car to remain a franchised Hyundai dealer.

This case appears to establish that besides the statutory protections dealers face, incentives that determine discretionary allocations are not discriminatory. This begs the question of how important discretionary allocations are as opposed to formula allocations. The answer appears to be *very*.⁶ Manufacturers tend to reserve “hot product” for their discretionary allocations; these are specialty or high-selling models and trims that are typically either not allocated at all in the standard allocation or are under-allocated. These discretionary allocations can constitute up to 15% of a dealer’s inventory. A public records request shows no formal complaints filed with the DMV regarding a dealership’s allocations during my sample period, reflecting that this is likely viewed as a settled matter. Beyond these anecdotal claims, I will show empirical evidence that assortment of trims is important to dealership profitability. When discussing the importance

⁶See “Allocation: It’s more than ‘Turn and Earn.’” *Bass Sox Mercer Law Firm*, October 9, 2019; “GM Discourages Dealers from Markups on Hot Vehicles.” *Motor Trend*, January 20, 2022

of trims in the rest of the paper, I will be doing so with the understanding that this is essentially a decision variable in the manufacturer’s control.

1.2 Economic Context

There are two countervailing economic forces and two corresponding strains of literature that are directly relevant to the question of downstream concentration in vertical relationships. The first is that of the logic of vertical restraints (Rey and Tirole 1986) and the classical issue of double marginalization (Tirole 1988). A more modern restatement of the problem in the context of vertical contracting can be found in the first two sections of Lee, Whinston, and Yurukoglu 2021. A manufacturer that must sell its product via a retailer who exercises market power experiences, in the simplest case, an efficiency loss. The retailer marks up above the wholesale price, and overall output is lower and price higher than if the manufacturer sold directly to consumers. This constitutes an efficiency loss from a total surplus perspective, as well as from the perspective of the manufacturer and consumer. The more concentrated the retail sector, the higher retailers can mark up over wholesale, and the worse the surplus loss.

The theory of vertical restraints developed as a way to explain the types of contractual relationships that were observed in practice, and it demonstrates how simple provisions can rectify the double marginalization problem. As discussed in Rey and Tirole 1986, Rey and Stiglitz 1995, Tirole 1988, and many others, quantity forcing or resale price maintenance can require the retailer to sell a quantity or at a price dictated by the manufacturer, allowing output and price to be set at the level that the manufacturer would prefer. In the simplest case these vertical restraints completely resolve the efficiency loss. One of critics’ primary concerns regarding the status quo of car retailing is that many conceivable vertical restraints are prohibited by law: quantity forcing and resale price controls are illegal, and to make matters worse, dealers are insulated from threats of entry, perpetuating concentration that could typically be ameliorated by competition from newcomers.

The other relevant strand of literature is that concerning the franchising relationship. Franchising is generally explored in the economics literature as a solution to the principal-agent problem that comes with delegating retailing to an entity besides the manufacturer (Lafontaine and Margaret E. Slade 1998). Primary concerns are relationship-specific investments and quality/effort on the part of the retailer: non-price decisions on the retailer’s part that affect consumer perceptions of the brand’s overall image and value. When intrabrand competition is too fierce, retailers lack incentive to make these investments. Murry 2018 has directly shown that this is an issue in car retailing. Using a credible instrumental variables strategy, he shows that dealers in more competitive markets advertise less, on average. The mechanism is that advertising generates a

negative externality from the perspective of the dealer: It not only benefits me, but also my competitor. In this case, downstream concentration can be a good thing, in that the consolidated dealer will spend closer to the “ideal” (from an efficiency perspective) amount on advertising.

Indeed with car retailing in mind, Martín-Herrán, Sigué, and Zaccour 2014 develop a simple model of a vertical relationship with spillovers from advertising and show that, for a plausible range of demand curves and levels of spillover, either the consolidated or unconsolidated downstream sector can be the most efficient. Hence, with the countervailing forces of pricing power and advertising spillover in mind, the ideal level of consolidation in the dealer network is an empirical question. This paper aims to provide the first empirical evidence on how this balance has played out in practice.

A handful of empirical studies have addressed other aspects of the car franchising relationship. Nurski and Verboven 2016 discuss an aspect present in Europe but not the United States; that is, the ability for manufacturers to enforce exclusive dealing contracts. They find that dealers obviously have an incentive to unilaterally deviate and sell additional makes on their own, but as a whole, the dealer network is more profitable under an exclusive dealing regime. Albuquerque and Bronnenberg 2012 address the impact of exit restrictions on dealer networks during the Global Financial Crisis. General Motors in bankruptcy chose to terminate hundreds of their dealers at great expense, partly borne by taxpayers as part of a much-publicized bailout. Albuquerque and Bronnenberg 2012 use individual-level transaction data from San Diego to estimate the operating costs and margins at the area dealerships and find that in this case, it may have been ostensibly justifiable, as it is profitable for GM to close a lower-performing dealer in the metro area. Murry 2018, as mentioned above, considers the externality that additional dealer effort may create through its advertising effort.

Related work in the context of vertical relationships has explored how mechanisms such as exclusive dealing may foreclose upstream manufacturers from using downstream distributors or retailers (Mathewson and Winter 1984, Sass 2005, Asker and Bar-Isaac 2014). There is mixed evidence on price effects because of exclusive dealing, with Sass 2005 and Chen and Shieh 2016 providing empirical evidence that exclusive contracts between beer manufacturers and distributors helps align incentives in an efficiency-enhancing way, similar to the role that advertising plays in my case. Meanwhile, Burgdorf 2021, for example, argues that the same relationship between manufacturers and distributors hamstrung a nascent craft brewing industry that emerged in the 2010s.

Exclusive territories, a particular form of restricted entry, face the same empirical uncertainty as exclusive dealing regarding incentive alignment versus downstream foreclosure. Verboven and Yontcheva 2024 examine

the effects of an entry restriction that is state-imposed, in the case of Latin notaries in Europe. They find that in this particular case, the level of entry is below socially optimal. With regards to state-restricted exit, Murry and Newberry 2022 exploit state-by-state variation in franchise termination laws to assess the impact of the regulation on the number of restaurants in several large franchise chains. Using the entry model of Bresnahan and Reiss 1991, they estimate that there are around 5% fewer establishments in states with laws limiting franchise contract terminations. They do not address the question of whether this would be a socially efficient level of entry as posed by Mankiw and Whinston 1986.

Taken together, the existing empirical evidence leaves open important questions about the extent to which dealerships are able to exercise and sustain market power. Zananone 2009 provides an legal and institutional analysis showing that even when vertical restraints are limited, manufacturers substitute toward other mechanisms, namely showroom, service, and staffing requirements. Consistent with this view, the evidence in Murry 2018 indicates that some degree of downstream protection or coordination may be necessary to sustain efficient investment. Observed margins on new vehicle sales are relatively modest on average per Brunson and Barnes 2020, on the order of a few hundred dollars, and much of the policy-oriented critique of dealer protections, including that in Lafontaine and Scott Morton 2010, was developed during the Great Recession when the salient issue was an inability to close failing dealerships. Whether these concerns generalize—to the world of the long bull market pre-pandemic and the increasing ownership concentration since the mid 2010s—remains an open empirical question that this paper tries to answer.

2 Data

2.1 Registrations

Data on car registrations come from the Texas Department of Motor Vehicles. It contains the universe of registrations in Texas from the beginning of 2016 to the end of 2019. Each registration contains the vehicle identification number (VIN) of the vehicle registered, the date the registration is effective, the zip code of the owner, the mileage of the vehicle, the type of registration (individual, business, government, etc.), and if it is sold by a dealership in Texas, the dealership’s general distinguishing number (GDN) and sale price of the vehicle. I decode VINs using the National Highway Traffic and Safety Administration database to obtain make, model, and trim information. New vehicle sales are not explicitly flagged and must be inferred, which I do by assuming that cars sold with less than 250 miles on the odometer and within one year of the car’s model year are new. Figure 1 justifies this choice, as the density of cars sold drops off significantly before 250 miles and the rate of cars that are surely used (greater than one year old) is increasing in vehicle

mileage. I focus only on sales to individuals in my analysis.

I restrict the analysis to models with over 10,000 new unit sales during the sample period. This restriction excludes models from some lower-volume and luxury manufacturers (notably omitting BMW), but keeps the bulk of the model lineup from the major domestic (Ford, Chevrolet, Dodge, Jeep, Ram) and international (Toyota, Nissan, Honda) manufacturers. To give an idea of the types of cars that are contained in this sample, Table 1 shows a count of the unique models I consider per manufacturer, split by NHTSA-decoded body type. There are many pickups, sedans, and SUVs that I consider, though notably fewer coupes and vans. Overall, with these restrictions I end up with 2,782,349 registrations in my sample, or roughly 750,000 sales per year. New car sales statistics in Texas for 2016–2019 were not immediately available, but the website F&I Tools estimates approximately 1.5 million new vehicles were sold in Texas in 2024.⁷ The restrictions on individual registrations and popular models likely constitute the bulk of the discrepancy with my analysis sample.

Because prices are negotiated, I see different prices being paid for the same car in my data. I compute different measures of price to measure different compositional changes I may see in my data. First I store mean price $\bar{P}_{it}$, which is the arithmetic mean of all of the cars that a dealership sells in a given period. Second, I compute a measure of markup that captures the percent deviation from trim-level average at which a dealer sells their cars:

$$\tilde{P}_{ijkt} := \frac{P_{ijkt} - \bar{P}_{jkt}}{\bar{P}_{jkt}}$$

where i indexes individual transactions, j indexes models, k indexes trims, t indexes model-year, and $\bar{P}_{jkt}$ is the average price of the year-model-trim combination across all dealers. Within a given period, I calculate and store the median markup at which a dealer sells their cars.

I choose to rank trims within a model-year by their average price, based on the assumption that trims are vertically differentiated and the relationship between price and consumer valuation is monotone. This assumption would fail, if, for example, supply of a “worse” trim was severely restricted relative to a “better” trim. What I call *trim rank* is the set of evenly spaced points on $[0, 1)$ for a given model-year. For N trims this would be $\{0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}\}$. I prefer this approach because it separates consumer valuation of trims from the price: Trim ranks should instead be thought of as quantiles of the distribution of trim valuations across consumers.

Finally, I calculate and store a measure I call “mean baseprice” for a dealer in a given period. For each model and model-year combination, I call $\bar{P}_{j0t}$ the base price of the model—that is, the average price of the

⁷See “New Car Sales by State” on factorywarrantylist.com

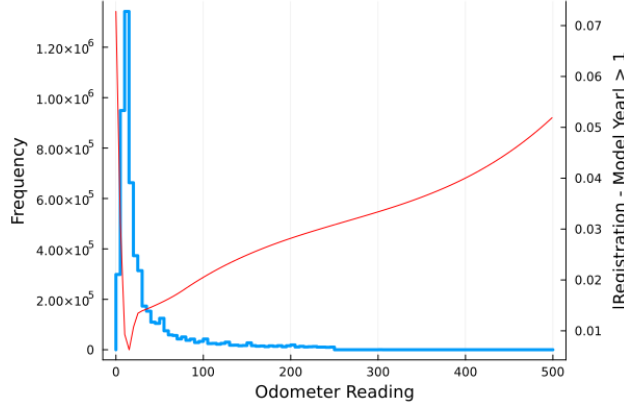


Figure 1: Histogram of vehicle odometer readings on left-hand axis, with rate of “definitely old” cars (cars whose model year is more than one year older than the registration date) on the right-hand axis. The distribution of odometer readings drops off sharply before 250 miles, and the rate of definitely old cars rises monotonically after an initial spike.

lowest rank trim across all dealers. This is a measure of the model mix that a dealer sells, such that if the average baseprice at which they are selling cars changes from period to period, that necessarily means that they are selling a different mix of models (as opposed to selling higher quality trims of the same model, or marking up the same vehicles a higher amount).

Trims as decoded by the NHTSA tend to be fairly granular. On average, each model has 3.77 different trims. I see more trims in my data than manufacturers advertise—for example, Chevrolet advertises four different trims for their Traverse SUV, but I see in my data as many as six depending on the model year. This arises because the NHTSA VIN decoder distinguishes accessory packages not always advertised by manufacturers—in the case of the Traverse, the “LT” trim that is advertised is further differentiated with “1LT” and “2LT” to distinguish a car with basic upholstery and infotainment system from a car with higher quality upholstery and infotainment as well as a power rear lift gate. I consider the fact that my data has more trim options as an advantage, as this gives me a more fine-grained distribution of trims to consider in my analysis. I perform basic string normalization to merge trims that differ by, e.g., punctuation, and I drop trims that sell less than ten units overall. Table 2 summarizes the average variation in prices within and across trim, by manufacturer. For manufacturers that sell many trims, trims tend to capture the majority of the price variation within models. For manufacturers such as Nissan, who offer very few trims per model, accounting for trims does little to reduce the variation in prices. It is likely that “dealer optional add-ons” account for more price variation in these manufacturers; however, in the last column of Table 2 it is clear that for the majority of dealers, trim information accounts for a significant amount of the variation in price within models.

	SUV/Crossover	Coupe	Sedan	Pickup	Van
CHEVROLET	6	1	2	2	0
CHRYSLER	0	0	0	0	1
DODGE	2	1	1	0	0
FORD	4	1	3	3	0
GMC	4	0	0	2	0
HONDA	3	0	2	0	1
HYUNDAI	2	0	2	0	0
JEEP	5	0	0	0	0
KIA	3	0	2	0	0
LEXUS	2	0	1	0	0
MAZDA	1	0	1	0	0
MERCEDES-BENZ	2	0	0	0	0
MINI	1	0	0	0	0
NISSAN	5	0	4	2	0
RAM	0	0	0	3	0
SUBARU	3	0	0	0	0
TOYOTA	3	0	3	2	1
VOLKSWAGEN	1	0	1	0	0

Table 1: Count of unique makes by manufacturer and vehicle type

2.2 Dealerships and Transactions

I have two pieces of data from the Department of Motor Vehicles on car dealerships. The first is a list, as of May 2022, of all of the people and entities holding dealers’ licenses in Texas. Each entry contains the name of the entity or person, a doing-business-as name if applicable, an address for the place of business, and a general distinguishing number (GDN) that acts as a license number to identify the dealer. This list encompasses franchise car dealers, independent car dealers, participants in the wholesale market, as well as RV and motorcycle dealers. In order to identify franchise dealerships, I match names of the dealerships with the makes in my analysis and known brand aliases, as most dealerships clearly indicate in their d/b/a name which makes they sell. I perform a back-check for missed dealerships by inspecting license numbers associated to at least fifty new car sales of a single make and find about one hundred false negatives. I re-inspect my list to catch false positives, of which there are a handful. (Take, e.g., Bob Ford Chevrolet, who sells Chevrolet but not Ford models.)

The second piece of data that I have is a list of all entries, exits, and transactions among dealerships from the beginning of 2017 to the beginning of 2024, which I will broadly call “events.” These data contain information on the entities that are transacting, entering, or exiting, as well as a relation of DBAs and GDNs that identify the dealerships affected, $R(DBA_{\text{buyer}}, GDN_{\text{buyer}}, DBA_{\text{seller}}, GDN_{\text{seller}})$. When dealerships are transacted, new GDNs are created, so there is not continuity in GDNs across owners. There is no notion of R being a functional mapping of $DBA_{\text{buyer}} \times GDN_{\text{buyer}} \rightarrow DBA_{\text{seller}} \times GDN_{\text{seller}}$, which creates issues

Make	Avg. Price	No. Trims	Std. Price Within	Std. Price Across	Within / Across
CHEVROLET	36448.95 (11244.23)	4.47 (1.32)	4244.43 (1755.09)	4942.6 (2295.44)	0.86
CHRYSLER	36369.72 (713.24)	4.84 (2.88)	4063.21 (2034.36)	3398.63 (2554.43)	1.2
DODGE	30872.43 (5687.19)	4.4 (1.9)	4698.94 (3389.1)	5526.39 (4960.79)	0.85
FORD	40035.1 (12925.97)	3.38 (1.94)	7190.09 (4718.9)	3972.55 (3206.02)	1.81
GMC	48161.33 (10738.5)	4.24 (1.8)	4846.22 (1527.82)	6108.58 (1443.07)	0.79
HONDA	28163.01 (5519.53)	7.85 (3.51)	2057.44 (313.41)	2926.43 (834.49)	0.7
HYUNDAI	22718.1 (4456.92)	2.87 (0.84)	2787.02 (630.24)	1503.01 (925.0)	1.85
JEEP	32519.31 (8274.99)	4.01 (1.82)	4183.5 (2393.34)	3012.59 (2198.61)	1.39
KIA	22359.09 (4193.51)	3.47 (1.14)	2705.59 (816.74)	2007.28 (1126.85)	1.35
LEXUS	45681.98 (4952.91)	1.44 (0.8)	4025.55 (687.3)	427.31 (755.17)	9.42
MAZDA	24745.58 (3456.05)	3.59 (0.91)	1843.49 (146.36)	2310.99 (507.24)	0.8
MERCEDES-BENZ	54967.73 (9178.41)	5.07 (1.65)	4831.36 (863.38)	8059.22 (3799.7)	0.6
MITSUBISHI	21495.21 (680.85)	2.33 (0.74)	2568.04 (282.85)	767.94 (236.83)	3.34
NISSAN	26629.71 (8003.33)	1.13 (0.7)	3977.28 (1457.1)	71.38 (391.98)	55.72
RAM	43258.82 (7297.11)	5.08 (2.38)	5436.33 (1477.39)	5107.48 (3291.78)	1.06
SUBARU	29963.33 (2886.69)	8.02 (1.34)	1662.75 (255.91)	2884.47 (546.66)	0.58
TOYOTA	31116.71 (7870.33)	4.53 (2.66)	3109.39 (990.03)	2496.58 (1605.79)	1.25
VOLKSWAGEN	22208.29 (3545.31)	5.32 (0.86)	2584.98 (368.38)	2681.61 (366.23)	0.96

Table 2: Average price, number of trims, and variation within and across trims for all manufacturers. Numbers in parentheses indicate standard error of these averages across models. All aggregates are sales-weighted at the appropriate granularity. For many manufacturers, trim captures a majority of the variation in price; less so with manufacturers like Nissan that tend to have few trims per model.

tracking dealerships over time. For example, two dealerships with different GDNs may be transacted, but they are not uniquely mapped to the new GDNs that are created. This issue occurs once in my data.

Coupled with the inherent multiplicity in my data—one dealership can have multiple GDNs, and multiple dealerships controlled by the same entity in the same municipality can have the same GDN—I have to accept a small degree of ambiguity tracking dealerships over time. I track GDNs associated to dealerships as sets, and where it is truly ambiguous for a particular registration where the customer bought their car, I default to assuming they bought from the dealership nearest to them. There are 65 dealerships in my initial May 2022 dataset that share GDNs with another dealership or that have multiple GDNs associated to their dealership.

My ultimate goal in this data cleaning exercise is to create a list of franchise dealerships in Texas, where each entry in the list represents a unique physical location where cars are sold. Furthermore, when comparing dealer performance over time, my goal is to create a way to track ownership changes and to “time travel” to a given point in time and to interrogate the state of the world to find the GDNs of, for example, nearby dealerships. That said, I treat two dealerships in the DMV data as a single dealership location when they have the same address; when they have the same DBA and are physically adjacent (less than one mile apart); or when they have the same GDN and are physically adjacent. This gives me a list of unique dealer locations. I count 1134 distinct dealers in my data using this method. One assurance that this is the correct granularity of merging is that where there are multiple makes sold at a given location (e.g., Jim’s Ford and Jim’s Mazda), I always see the entire location transacted.

The last step to prepare my *events dataset* for analysis is to compile a set of “diffs” to track changes over time. Starting with the last event prior to the date of my list of dealers in May 2022, I work backwards and make a best-effort attempt to apply the changes to the state of the world implied by the event (accepting ambiguity in the one case where it is necessary). By working backwards to the beginning of my data and collecting these diffs as I go, understanding the state of the world at a given point in time is as simple as applying a set of mutations to a piece of in-memory state. This is deterministic and essentially costless in terms of computational complexity. Ultimately, I use a convenience function written in Julia:

```
timetravel(e) do
    [code in e's state of the world...]
end
```

that applies the diffs to rewind back to the point just before (or after) the event e and executes the code supplied “in the past.”

Dealer groups often incorporate separate entities for each dealership they own, and the names of these

entities may not bear resemblance to each other. For example, the AutoNation group incorporates entities with names similar to their dealership d/b/a names (think “Galveston Honda, LLC” and “Corpus Christi Ford, LLC”), hence without ownership tracing it is impossible to discern they are related. In order to reconstruct the set of holdings for each dealer group, data from OpenCorporates⁸ are used to find each registered entity and identify the boards of directors and controlling entities. These data are the most complete set of board membership and joint ownership data available, enabling the tracking of entities across state and even national boundaries. Where entities share a mutual controlling entity or at least one member of their boards of directors, I assume that the entities are commonly owned. I manually confirmed the effectiveness of this strategy with several of the largest known dealer groups, and this approach was able to link all the dealerships that these dealer groups hold in Texas. Figure 2 gives a histogram of the size of each dealer group. Most dealer groups are small and hold fewer than ten franchise dealerships, but there are outliers that hold as many as several dozen franchise dealerships in Texas as of May 2022. I calculate that the 1134 distinct dealerships in Texas are owned by 577 dealer groups as of May 2022. The largest four dealer groups account for 10.49% of all the dealerships in Texas.

In order for me to consider a transaction as part of my analysis sample, I require a year’s worth of registration data from the dealership before and after the transaction. This narrows the number of transactions in my sample to 38 transactions involving 44 dealerships between June 7, 2017 and March 21, 2019. Table 3 gives a breakdown of dealership locations and transaction locations, as well as the CR-4 concentration within each metro (the share of dealerships held by the four largest groups). Despite larger dealer groups tending to locate in the larger metros, there is still less concentration than in smaller metros like Austin, Corpus Christi, or El Paso. In these smaller metros, one large group can more easily dominate the market. There are also more transactions in large metros, with some exceptions to that rule. Figure 3 shows the size of the dealer groups buying and selling the dealerships in my analysis sample. Most of the transaction volume appears to come from smaller dealer groups, again challenging the narrative of dealer roll-ups. Where large dealer groups transact dealerships, they appear not to have another large dealer group as counter-party. Note there are a number of dealers that are “exiting the market”—that is, they are selling the only dealership(s) they own.

2.3 Measures of Market Power and Concentration

Market power is a local phenomenon in the case of car dealerships. A standard practice in the industry is to divide geographic space into “exclusive territories” and concern oneself with the potential demand

⁸OpenCorporates, Limited. See <https://opencorporates.com/>

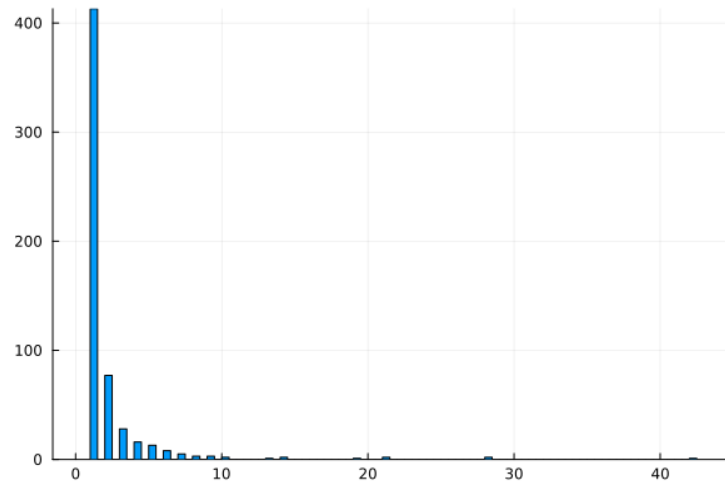


Figure 2: Histogram of dealer group sizes in Texas, as of May 2022.

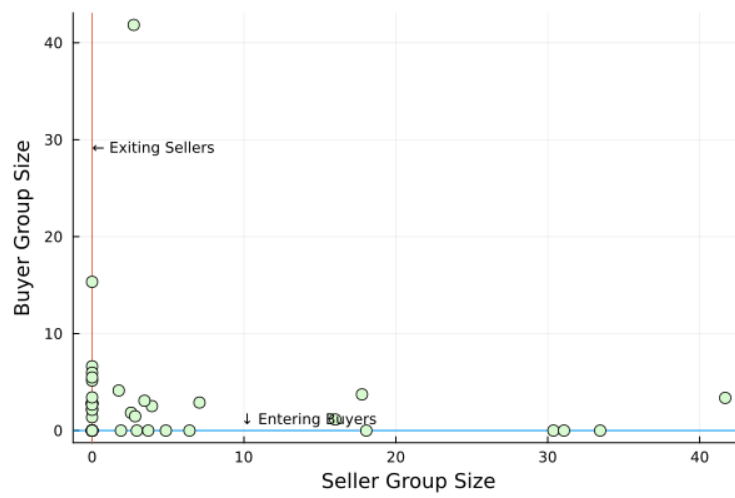


Figure 3: Size of dealer groups buying and selling a given dealership. Size of the selling group is along the x-axis, and size of the buying group along the y-axis

MSA	Dealerships	Transactions	CR4 (Num. Dealerships)
OUTSIDE MSA	243	11	0.099
Dallas	213	10	0.197
Houston	183	5	0.191
San Antonio	81	2	0.272
Austin	64	4	0.312
Beaumont	28	0	0.429
Corpus Christi	26	3	0.538
El Paso	26	0	0.654
Longview	25	3	0.280
McAllen	24	0	0.500
Killeen	22	0	0.455
Waco	21	0	0.571
Lubbock	20	2	0.650
Sherman	18	2	0.556
Brownsville	17	0	0.588
Amarillo	16	0	0.438
College Station	14	0	0.500
Abilene	14	0	0.571
Tyler	13	0	0.769
Midland	12	0	0.417
Wichita Falls	12	0	0.583
Victoria	11	0	0.818
Laredo	11	0	0.909
Odessa	10	1	0.800
San Angelo	8	2	0.750
Eagle Pass	2	0	1.000

Table 3: Table of number of dealerships, number of transactions, and CR-4 concentration ratio in number of dealerships for each metro area.

of the customer base in the exclusive territory and the possibility of “poaching” consumers from nearby dealers’ exclusive territories (or the risk of one’s own customers getting poached). Per that description, I find that 44.77% of buyers come from a dealer’s exclusive territory, calculated by creating a Voronoi tessellation of dealer locations on a make-by-make basis. A standard economic practice would then be to segment geographic markets at the MSA level or a similar political designation and calculate market share within those boundaries. I find this to be inadequate for the retail car market in Texas for two reasons. First, per Table 3 almost a quarter of dealers in Texas lie outside the bounds of an MSA. Second, consumers often cross market boundaries to buy cars. Of those that did *not* buy from the closest dealer, consumers traveled an average of 40.37 miles from their home zip code to their chosen dealership, and 9.54% traveled more than 100 miles. These are distances that defy simple geographic segmentation, so I prefer to consider the state as a *single* geographic market and calculate local market share measures using kernel distance weighting.

The compromise that I use when calculating market shares is to capture rough measures of differentiation based on distance and body type. The former is as discussed, and the latter is to account for the fact that

makes that do not sell any pickup trucks, e.g. Mazda, are not substitutes for makes like Ford that sell primarily pickup trucks. To calculate the market share of a dealer i in period t , I proceed in the following manner:

1. Calculate proportion of sales by body type for a particular dealer. If q_{ict} represents sales made by dealer i in period t of body type c (mnemonic “vehicle class”), and $\mathcal{C}$ is the set of all body types, define

$$\pi_{ict} \equiv q_{ict} / \sum_{c' \in \mathcal{C}} q_{ic't}$$

2. Within a body type c , calculate a weighted share of sales *of the entire dealer group* against dealers within a radius R using weight function w :

$$s_{ict} = \frac{\sum_{j \in g_i} w(d^{geo}(i, j)) * q_{jct}}{\sum_{k \in \mathcal{R}_i} w(d^{geo}(i, k)) * q_{kct}}$$

Here the set g_i is the set of all nearby dealers that are in the same dealer group as i . I define $\mathcal{R}_i = \{j : d^{geo}(i, j) < R\}$. The function $d^{geo}(i, j)$ represent the distance in miles between dealers i and j , and $w : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $w(0) = 1$ and $w' \leq 0$. The weight function I choose for my specification is the linearly decreasing weight with maximum radius of 50 miles:

$$w(x) = \left(1 - \frac{x}{50}\right) * 1\{x < 50\}$$

3. Calculate overall market share by taking a weighted average of body type shares:

$$s_{it} = \sum_{c \in \mathcal{C}} \pi_{ict} s_{ict}$$

HHI calculation proceeds in an analogous manner. For each dealer group g within radius R I calculate their market share (from the perspective of i) as

$$s_{gct} = \frac{\sum_{j \in g} w(d^{geo}(i, j)) * q_{jct}}{\sum_{k \in \mathcal{R}_i} w(d^{geo}(i, k)) * q_{kct}}$$

The overall (local) HHI from the perspective of dealer i is

$$HHI_{it} = 10,000 * \sum_{c \in \mathcal{C}} \left[\pi_{ict} \times \sum_{g \in \mathcal{G}} s_{gct}^2 \right]$$

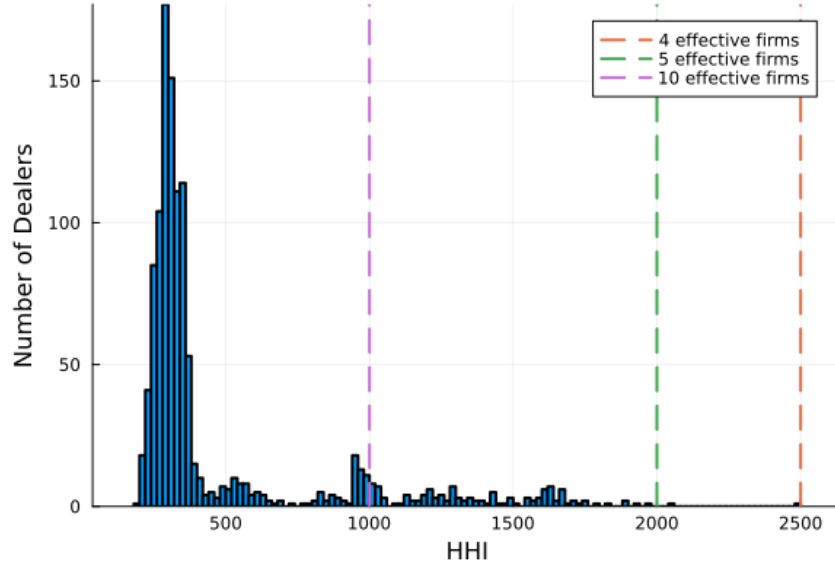


Figure 4: Cross-sectional measure of local HHI across all dealers in Texas, calculated for the first quarter of 2018 using the procedure outlined above.

where $\mathcal{G}$ represents the set of all dealer groups within R miles of i .

Given this approach to measuring concentration, there is wide cross-sectional variation in the amount of market concentration in Texas. Figure 4 shows the local measure of HHI calculated at every dealership in Texas for sales in the first quarter of 2018. A handful of dealers exhibit strong market power with few rival dealer groups (or else poor substitutes) nearby. However, the vast majority exhibit a healthy amount of competition, with a majority facing more than ten effective firms using the symmetric Cournot measure ($HHI = 10,000/N$ implies N symmetric Cournot firms). This falls below the cutoff of 1,800 that the Department of Justice’s 2023 Horizontal Merger Guidelines categorizes as a concentrated market.

Figure 5 constructs a more global measure of HHI by setting $w(x) = 1$, $R = \infty$ and using body type weights π_{ict} from the most representative dealer possible. For every quarter in my sample, I recalculate this global HHI measure and plot the time series in Figure 5. The level is not as important as the trend: Concentration is indeed increasing, and over the course of two and a half years it increases by roughly 50%.

The transactions in my data are not necessarily all concentration-increasing, however. From the perspective of the dealership being transacted, it may be joining a dealer group that is locally stronger or locally weaker. To calculate how a dealer’s local market share changes as the result of a transaction, I perform a counterfactual exercise using pre-transaction sales data. Using data from the third and fourth quarters prior to the transaction, I calculate a dealer’s market share, first assuming that it is part of the selling group’s portfolio (as was actually the case), and then assuming that it is part of the buying group’s portfolio (a

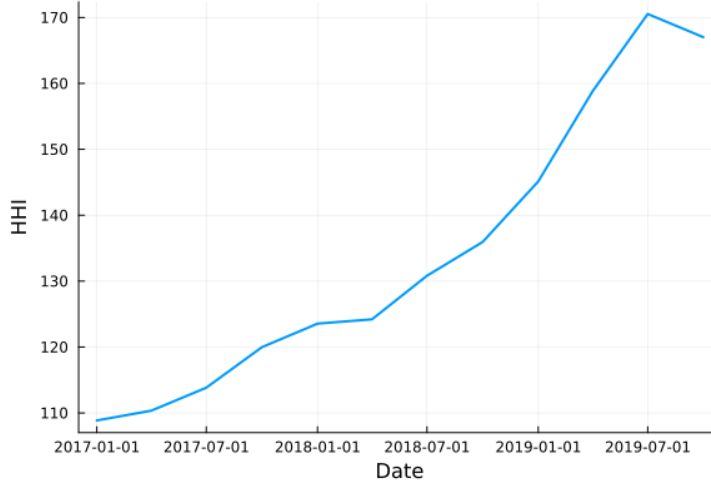


Figure 5: Time series of global HHI measure across my sample frame. Concentration is increasing over the course of my sample.

counterfactual case). The difference between the two is what I call the local market share gained or lost as the result of a transaction.

For nine dealerships involved in transactions, there is no change in market share. This is primarily because the parties involved are single owner-operators, however there is also one case of a Houston-based group and a Dallas-based group transacting an Austin dealership. The other 35 dealerships experienced some change in market share as the result of being transacted. Figure 6 shows a histogram of the market share changes that these dealerships experienced as a result of being transacted. Despite the overall trend of increasing global concentration, it is roughly equally common to increase market share as it is to decrease, and the magnitudes of these changes can be large, on the order of $\pm 10\%$ or more.

3 Analysis

For each dealership that is transacted, I analyze its effect using a year's worth of registrations before and after the transaction. For each quarter of data, I aggregate the registrations into a set of outcomes—I am mainly concerned about total sales, units sold, components of price (mean price, median markup over average, mean base price), average trim rank, and the number of trims sold. This aggregation leaves me with a balanced panel of one outcome per dealership per period. To estimate the treatment effect of a particular transaction, I collect a set of dealerships that are plausibly unaffected by the transaction. I determine this control group to be dealerships that sell the same makes as the treated dealership; dealerships that are more than 50 miles away from the treated dealership; dealerships that are not also owned by any party to the transaction; and

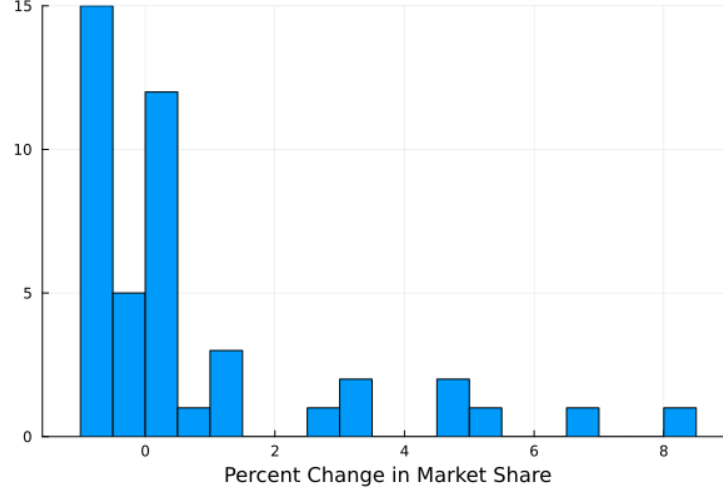


Figure 6: Histogram of market share changes for dealerships that experienced a non-zero change in their local share by changing dealer groups

dealerships that are never transacted, enter, or exit themselves.

Given the fact that there are multiple transactions in progress at a given time, this control dealer restriction is fairly conservative. Treated dealerships in larger metro areas tend to restrict the possible set of control dealers. As a compromise, I use a dynamic radius that can shrink to less than 50 miles when there are more than three other dealers of the same make in this range. Hence, the closest control dealerships are either the fourth-closest to the treated dealer or are 50 miles away, whichever is smaller. On average, treated dealerships have 27 control dealerships to compare against. Despite this relaxation to allow more urban dealerships to function as controls, there are still measurable demographic differences between treated and control groups. Figure 7 uses the demographics of nearby zip codes and the same distance weighting as discussed in the previous section to create a local measure of several different demographic variables for each dealership, and then plots the difference between treated and control dealers. Treated dealers tend to be more urban and higher-income than control dealerships. My preference to address this issue is to use a synthetic weighting in my regression specification, which I discuss below.

The analytical framework that I use to assess treatment effects is the event-study design. This approach compares differences between treated and control dealers within a set window before and after a transaction. At its most basic, this amounts to a two-way fixed effects regression of the form:

$$Y_{it} = \alpha_i + \delta_{m(i),t} + \sum_{\ell=-4}^3 \gamma_{\ell} D_i 1\{\tau_{it} = \ell\} + \varepsilon_{it}$$

where Y_{it} is unit i 's outcome in period t , α_i is a unit-specific fixed effect, $\delta_{m(i),t}$ is a time fixed effect that I

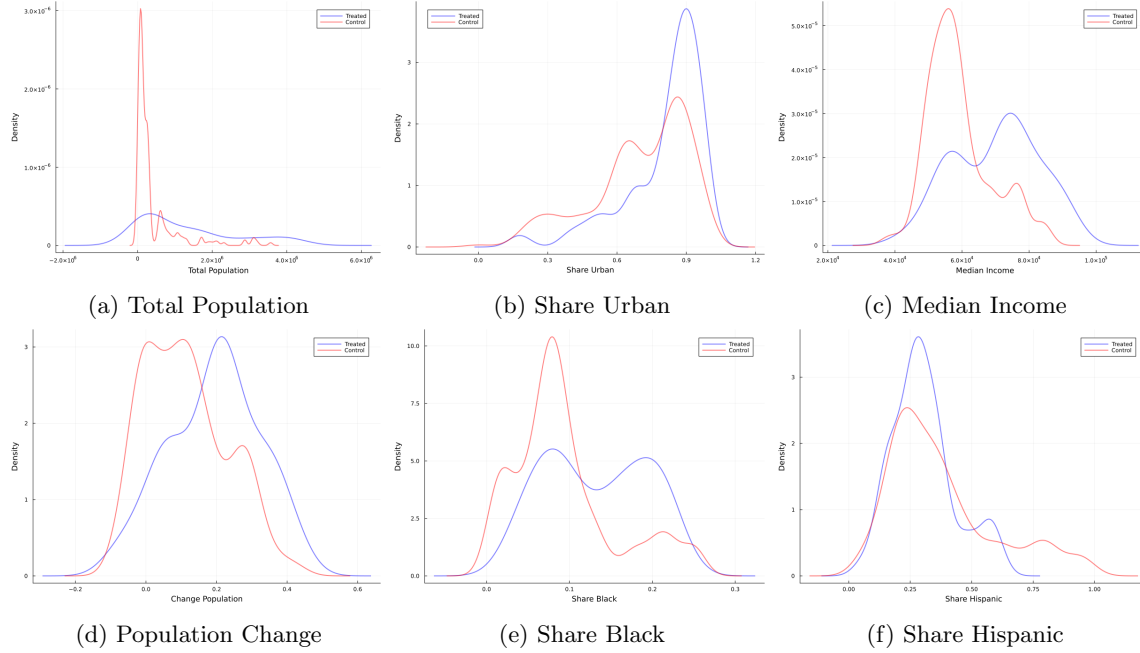


Figure 7: Differences between treated (blue) and control (red) dealerships in local measures of consumer demographics. Control dealers tend to be less urban, less populous, and lower income than treated dealers.

also choose to have vary by manufacturer, D_i is a treatment indicator, and τ_{it} indicates event time. Event time τ_{it} indicates how many quarters before or after the transaction a period t . If a dealer i is treated at time T_i , τ_{it} is defined mechanically as $\tau_{it} = T_i - t$. In my case, $\tau_{it} = 0$ is the quarter in which the transaction takes place.

I try several regression estimators of this event study for robustness. The first is the simple two-way fixed effects specification above. As is well known, I must normalize $\gamma_{-1} = 0$ because of colinearity. The second is the cohort-specific event study estimator from Sun and Abraham 2021. They suggest accounting for treatment effect dynamics by estimating cohort-specific effects. In my case, cohorts are defined by quarters in which transactions took place:

$$Y_{it} = \alpha_i + \delta_{m(i),t} + \sum_{\ell=-4}^3 \sum_{g \in \mathcal{G}} \gamma_{\ell}^g D_i 1\{\tau_{it} = \ell \wedge i \in g\} + \varepsilon_{it}$$

Here, $\mathcal{G}$ represents the set of cohorts, with g a representative element. After estimating cohort-specific parameters γ_{ℓ}^g , they aggregate with a weighted average to arrive at an overall effect estimate γ_{ℓ} , with the weights being proportional to the number of treated dealers in each cohort.

My last and preferred approach is to mimic the synthetic control approach of Brand, Garmon, and Rosenbaum 2023. For each treated dealer, I assign a set of control dealers with regression weights that sum to

one. I then estimate the two-way fixed effects estimator as above using these regression weights, with the only difference being that the fixed effect α_i is specific to the group of treated and control dealers. I create regression weights for control dealers in my regression by weighting based on non-outcome covariates. In particular, I use demographic variables total population, population change, median income, share white, black, and Hispanic. These variables capture some measure of potential demand, and I calculate them for each dealership by aggregating said variables from nearby zip codes using Census and ACS data from 2020. The population change is assessed as the percentage change in a zip code’s population between the 2010 and 2020 Censuses. I use the same distance weight as in calculating market share—for population, that is

$$w^{pop}(z, i) = \left(1 - \frac{d^{geo}(z, i)}{50}\right) * 1[d^{geo}(z, i) \leq 50]$$

where z indexes zip codes, i indexes dealers, $d^{geo}(z, i)$ is distance in miles between a zip code and dealer, and $1[\cdot]$ is the indicator function. For the other demographic variables, for which I have averages at the zip code level, I must also weight by population:

$$w^x(z, i) \propto pop_z * w^{pop}(z, i)$$

for a given demographic variable x in a zip code z with population pop_z . With these six covariates in hand, I construct a distance metric to measure similarity of two dealerships in covariate space. The natural distance metric is the Mahalanobis distance:

$$d^{mah}(i, j) = \sqrt{(X_i - X_j)' V_X^{-1} (X_i - X_j)}$$

where i and j index dealers, X_i and X_j are column vectors of covariates associated with i and j , and V_X is the variance of the six measures across the set of all dealers. The goal of this approach is to weight control dealers such that they create a “synthetic” control to the treatment. Rather than assign equal weights to every observation and try to demonstrate covariate balance (it is not expected that all control dealers be similar to the treated dealer—Texas is a demographically diverse state), Brand, Garmon, and Rosenbaum 2023 and I take inspiration from Arkhangelsky et al. 2021 and set the regression weights of the control dealers such that they sum to one, with the regression weight of the treated dealer also set to one. I use Gaussian weighting of the Mahalanobis distance as regression weights, $w^{reg}(j) \propto \phi(d^{mah}(i_{treated}, j))$, where $\phi(\cdot)$ is the standard Normal density. This regression specification puts more weight on control dealers closer to the treated dealer in covariate space. In addition, my unit fixed effect α_i in this specification instead acts as a fixed effect for the entire treated-control grouping.

I estimate a separate regression for each of the outcomes of interest. In an event study setup where the first period pre-treatment is omitted and γ_{-1} is set to zero, the coefficient γ_j can be interpreted as

$$\gamma_k = (E[Y_{i,t+k} | D_{i,t+k} = 1] - E[Y_{i,t+k} | D_{i,t+k} = 0]) - (E[Y_{i,t-1} | D_{i,t-1} = 1] - E[Y_{i,t-1} | D_{i,t-1} = 0])$$

Hence, coefficient tests are tests that the difference between treated and control units is statistically the same as that in the last period pre-transaction. I claim another interesting quantity to test is $\gamma_{\tau_{post}} - \gamma_{\tau_{pre}}$ for some $\tau_{post} \gg 0$ and $\tau_{pre} \ll 0$. This can be interpreted as

$$\gamma_{\tau_{post}} - \gamma_{\tau_{pre}} = (E[Y_{i,\tau_{post}} | D_{i,\tau_{post}} = 1] - E[Y_{i,\tau_{post}} | D_{i,\tau_{post}} = 0]) - \quad (1)$$

$$(E[Y_{i,\tau_{pre}} | D_{i,\tau_{pre}} = 1] - E[Y_{i,\tau_{pre}} | D_{i,\tau_{pre}} = 0]) \quad (2)$$

Assuming there is some anticipation pre-treatment or some adjustment period post-treatment, this constitutes a “donut hole” difference-in-difference estimate of the treatment effect of the transaction. The same notion extends to averages of coefficients pre and post transaction. The ex-post difference-in-difference estimator that I construct and report from my event study regressions leaves a six-month hole on either side of the transaction:

$$H_0^{DID} : \frac{\gamma_2 + \gamma_3}{2} - \frac{\gamma_{-4} + \gamma_{-3}}{2} = 0$$

In order to relate different outcomes to each other for the same dealership, I create dealership-specific differences by differencing outcomes from their model-predicted control outcome. I construct this difference Δ_i^Y as

$$\Delta_i^Y \equiv (Y_{i,\tau_{post}} - E[Y_{i,\tau_{post}} | D_{i,\tau_{post}} = 0]) - (Y_{i,\tau_{pre}} - E[Y_{i,\tau_{pre}} | D_{i,\tau_{pre}} = 0]) \quad (3)$$

$$= (\gamma_{\tau_{post}} - \gamma_{\tau_{pre}}) + (\varepsilon_{i,\tau_{post}} - \varepsilon_{i,\tau_{pre}}) \quad (4)$$

In expectation, this value Δ_i^Y is the difference-in-difference estimate derived above, but with residual error that I can correlate across outcomes. This allows me to explore how dealers that tend to have increased sales also tend to have changes in other outcomes. Besides performing regressions for the treated dealerships, I also consider other dealerships that might be affected by the transaction. For each other dealership in the selling group’s portfolio, I perform the same event study regression, creating yet another set of plausible control dealerships by extending my restrictions in a straightforward way. These control dealerships are at

least 50 miles away from both the focal dealership and the transacted dealership, sell the same models as the focal dealership, are not held by a party to the transaction, and are themselves never transacted. I do the same for other dealerships in the buyer’s portfolio, and I do the same for the closest competitors to the transacted dealer, i.e. the three closest dealerships that sell the same make of car. Using the Δ_i^Y constructed for their associated treated dealership, I can test, for example, whether sales increases at a treated dealership spill over to nearby dealerships, or vice versa. I can get at this question by allowing the estimates of γ_ℓ to vary by Δ_i^Y . In my regressions I use spline basis functions ϕ_k to flexibly estimate this effect:

$$Y_{jt} = \alpha_j + \delta_{m(j),t} + \sum_{\ell=-4}^3 \sum_{k=1}^K \gamma_\ell^k D_{it} \{ \tau_{it} = \ell \} \phi_k(\Delta_i^Y) + \varepsilon_{jt} \quad (5)$$

where j is a dealer associated to treated dealer i . Now, the difference-in-difference estimate is local to a given value of Δ_i^Y . The relevant hypothesis test is whether the difference-in-difference estimate increases as Δ_i^Y increases. I accomplish this in practice by testing differences in two very close points. For $x' > x$ that are sufficiently close, the hypothesis test in vector notation is

$$H_0^{DERIV}(x) : (\phi(x') - \phi(x)) \cdot (\gamma_{\tau_{post}} - \gamma_{\tau_{pre}}) = 0$$

Note that this hypothesis test is local to x , and that as $x' \rightarrow x$ this is a test that the derivative of the difference-in-difference estimate with respect to x is zero. I only implement this particular test for my TWFE and synthetic control estimator: the Sun and Abraham 2021 specification becomes too saturated at this point. I also implement and test an average derivative that uses the empirical measure of the test statistic from $H_0^{DERIV}(x)$ in my data:

$$H_0^{DERIV} : \int (\phi(x + \varepsilon) - \phi(x)) \cdot (\gamma_{\tau_{post}} - \gamma_{\tau_{pre}}) dF_N(x) = 0 \quad (6)$$

All of the hypothesis tests in this section are linear transformations of regression parameters. I cluster regression standard errors at the unit level in all specifications and implement these hypothesis tests as exact linear transformations of the cluster-robust variance matrix of the estimated parameters.

4 Results

The first results that I present are the event studies of prices and quantities for transacted dealers, split by whether the transaction is market share-increasing or share-decreasing. Figure 8 shows the event study for

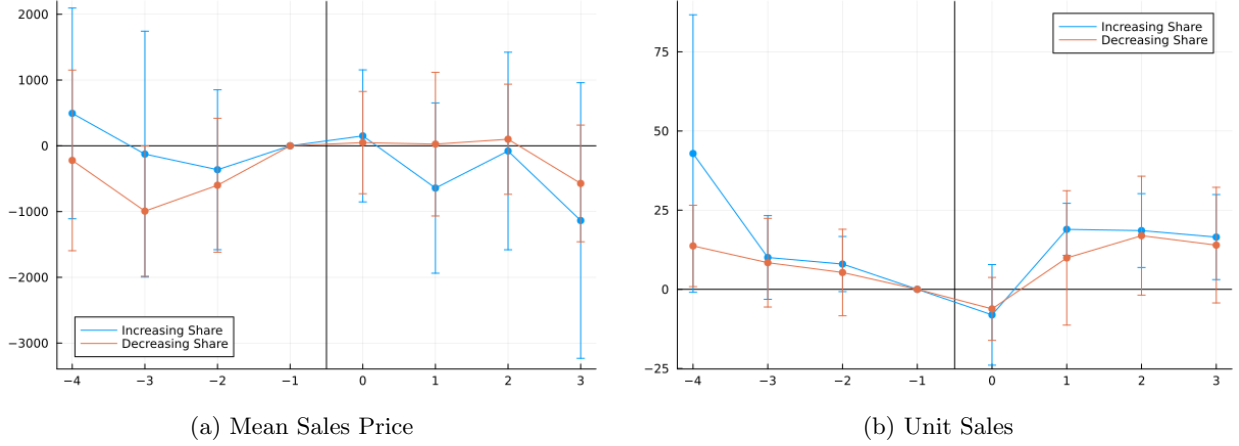


Figure 8: Event study results for transacted dealers using the Sun and Abraham regression specification. Results are split by whether the transaction was market share-increasing or share-decreasing.

price and quantity using the Sun and Abraham 2021 specification. There is no statistically significant effect on price or quantity throughout the entirety of the event study window, neither for the share-increasing nor the share-decreasing groups. Table 4 summarizes this more succinctly: Across the three specifications and for the two groups, I present donut hole difference-in-differences estimates of the effect of the transaction on all of my outcomes of interest. The only statistically significant outcomes are that the number of trims increase and markups decrease for transactions with increasing market share. One possible explanation for this effect is that groups with more capacity across their set of dealerships (or else better relational capital with regional managers) may be able to obtain a broader mix of cars than did the smaller group that is selling. As alluded to in the legal context section regarding GM’s allocation practices, these dealers may be lowering markups in return for this favorable allocation from the manufacturer.

In order to increase the size of the sample available to me, I perform the same exercise, but with dealers affiliated with the transacted dealers. Nearby dealerships that are part of the acquiring group will see an increase in market share as the result of a transaction, while dealerships that are part of the selling group will experience a decrease in market share. As in Equations 5 and 6, I will allow the diff-in-diff estimate for affiliated dealerships to vary by how much market share they gain or lose. Figure 9 plots the change in market share on the x axis and the diff-in-diff estimate for the change in price or quantity on the y axis. Here, I also include shading to indicate values of x where $H_0^{DERIV}(x)$ can be rejected for one alternative or the other (green being positive and red negative). Once again, I cannot find an effect on the price or quantity of a concentration-changing transaction. The average derivative of the diff-in-diff estimate with respect to market share is exceedingly modest and is not statistically significant for either outcome. Figure 13 performs the same analysis for other outcomes and finds no statistically significant average effect.

Outcome	TWFE	Synthetic Control	Sun-Abraham
Increasing Market Share			
Mean Baseprice	771 (472)	951 (557)	752 (421)
Mean Price	-865 (800)	-755 (795)	-791 (695)
Median Markup	-0.0204*** (0.00953)	-0.0173*** (0.00625)	-0.0181*** (0.00814)
Total Sales	1.02e+05 (5.68e+05)	9.79e+04 (3.48e+05)	-5.03e+04 (5.19e+05)
Total Units	-3.69 (17.8)	-2.48 (10)	-8.95 (16.8)
Trim Rank	-0.0203 (0.0175)	-0.0258 (0.0146)	-0.0201 (0.011)
Unique Trims	2.51 (1.3)	2.01 (1.2)	2.37*** (1.08)
Decreasing Market Share			
Mean Baseprice	-46.1 (586)	158 (511)	145 (392)
Mean Price	185 (870)	437 (694)	371 (431)
Median Markup	-0.00572 (0.00845)	-0.00328 (0.00844)	-0.000465 (0.00564)
Total Sales	1.2e+05 (4.21e+05)	9.07e+03 (2.87e+05)	1.94e+05 (3.73e+05)
Total Units	2.18 (12.7)	-0.499 (8.59)	4.39 (10.9)
Trim Rank	0.0258 (0.0275)	0.0281 (0.0189)	0.0132 (0.0155)
Unique Trims	2.28 (1.17)	1.85*** (0.909)	1.92*** (0.818)

Table 4: Donut hole diff-in-diff estimates of effect of transaction, for various outcomes and regression specifications.

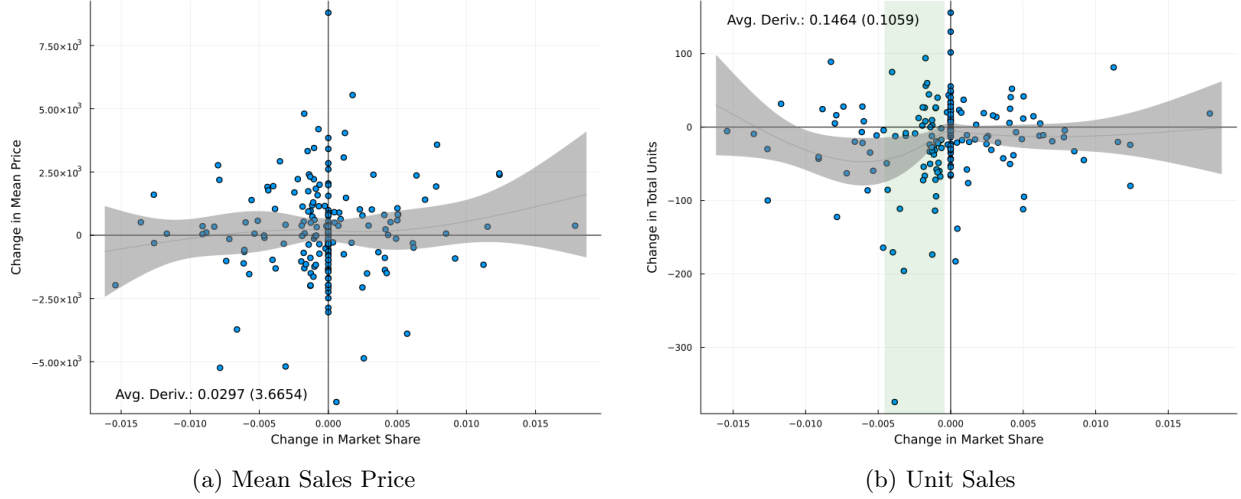


Figure 9: Difference-in-difference effects on mean price and quantity for affiliated dealerships, given the change in market share they experience because of a transaction. Average derivatives are positive but insignificant.

Although there were no major headline results, the transaction still produced meaningful correlations between the resulting outcomes. Table 5 presents measures of linear correlation between the residual differences of different outcomes as in Equation 3. I perform a hypothesis test against a null of zero correlation using the Student's t approximation to the sampling distribution.

Two mechanical results are that unit sales and total revenue are almost perfectly correlated, and that the different constituents of price (mean price, median markup, mean base price) are all significantly correlated. However, mean price and unit sales have no significant correlation. Moreover, there is a strong statistically significant relationship between the number of unique trims sold and unit sales. Dealers that tended to sell a wider variety of trims after a transaction tended to sell more cars. Dealers that sold higher trim cars sold at a higher average price, virtually by necessity, but they did not mark their vehicles up or down when doing so. Dealers who marked their cars up sold fewer units. The fact that markup can be correlated with sales while price is not suggests that there are significant compositional effects during a transaction that end up obfuscating the final effect on price, whereas markup is a more targeted measure able to more plausibly identify that the law of demand still holds.

The last set of results I present are related to spillovers of sales increases across dealers. The event study design that I estimate takes nearby dealers as the treated group, constructs a relevant control group, and

	Total Units	Total Sales	Median Markup	Unique Trims	Mean Price	Trim Rank
Mean Baseprice	0.12	0.127	-0.132	0.006	0.634***	0.083
Total Units	—	0.976***	-0.347***	0.462***	-0.122	-0.177
Total Sales	—	—	-0.264	0.512***	-0.079	-0.187
Median Markup	—	—	—	-0.214	0.358***	0.002
Unique Trims	—	—	—	—	-0.022	-0.009
Mean Price	—	—	—	—	—	0.58***

Table 5: Correlated Residuals across Outcomes

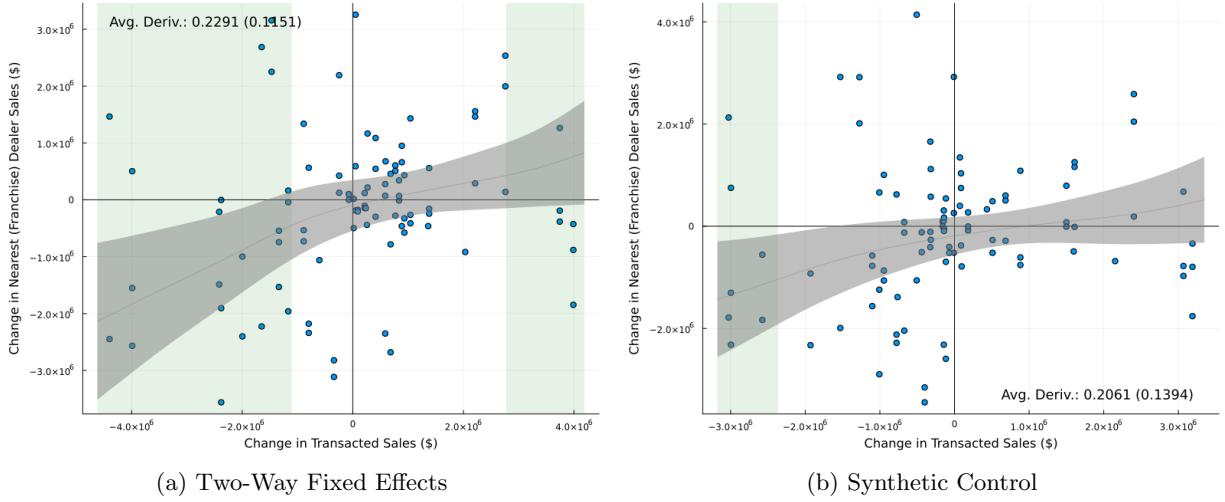


Figure 10: Effect of dollar revenue change of transacted dealer on nearest dealer of the same make. Estimate and derivative computed using kernel estimator of log distance at a given level of revenue change.

allows the event study parameters to vary by distance and change in sales at the transacted dealer:

$$Y_{jt} = \alpha_j + \delta_{m(j),t} + \sum_{\ell=-4}^3 \gamma_{\ell}(i) D_j 1\{\tau_{jt} = \ell\} + \varepsilon_{jt}$$

$$\gamma_{\ell}(i) = \gamma_{\ell}^0 + \gamma_{\ell}^1 \Delta_i^Y + \gamma_{\ell}^2 \ln d^{geo}(i, j) + \gamma_{\ell}^3 \Delta_i^Y * \ln d^{geo}(i, j)$$

Here, Δ_i^Y is the residualized change in sales at the nearby transacted dealer and $d^{geo}(i, j)$ is the distance in miles between the dealer in question and the transacted dealer. I do this exercise for the three closest (non-jointly owned) dealers that sell the same make as the transacted dealer, and as a placebo I run this regression for the three closest (non-jointly owned) dealers of any make. Figure 10 shows the result of the regression for dealers of the same make, and Figure 11 the result for dealers of any make. Depending on the specification, nearby franchise dealers exhibit a mild to statistically significant effect, with an average derivative around 0.2—that is, a revenue increase of \$100 at a transacted dealership is associated with a \$20 revenue increase at a nearby dealer of the same make. On the other hand, the effect does not spill over to dealers that do not sell the same make—the derivative is everywhere statistically zero.

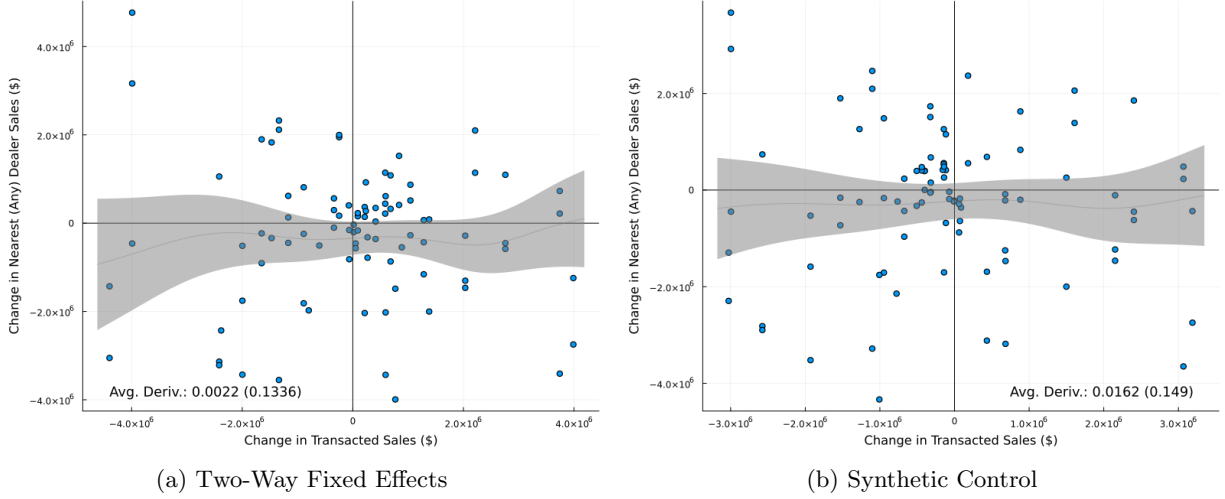


Figure 11: Effect of dollar revenue change of transacted dealer on nearest dealer of the any make. Estimate and derivative computed using kernel estimator of log distance at a given level of revenue change.

5 Discussion

In the previous section, I showed that despite large market share changes as a result of dealerships being transacted, there is no aggregate effect on price or sales volume. Dealerships that joined a locally more powerful dealer group sold a wider variety of trims at a lower markup. In the cross-section, dealerships that sold a wider variety of trims post-transaction also increased their sales volume. Dealers who increased their sales volume post-transaction had a positive spillover effect to nearby dealers, but only dealers of the same franchise.

I claim that this is indirect evidence of the allocations mechanism working effectively to discipline the market power effect from dealership transactions. For one, prices and sales volume did not change despite local market power changes on the order of 10% or more in either direction. The exact assortment of trims doled out as part of the allocation process appears to be vital to dealer profitability, as there is a strong and statistically significant correlation between dealers who sold an increased assortment of trims and dealers who increased sales volume. There is also evidence that dealerships that joined more powerful dealer groups and received a “better” assortment of trims via allocations sold those cars at a lower markup, aligning with industry sources reporting that manufacturers expect dealers rewarded with favorable allocations to sell at a more modest margin.⁹

Furthermore, direct evidence of the spillover effect of sales increases shows that the principal-agent dynamic of franchising theory is imminently applicable to car franchising. Evidence from Murry 2018 would suggest

⁹c.f. “GM Discourages Dealers from Markups on Hot Vehicles.” *Motor Trend*, January 20, 2022

that advertising is a particular channel through which dealer effort can have a positive spillover to neighbors, though I remain more agnostic and only claim that whatever the mechanism, the result of dealers’ efforts does diffuse to nearby competitors. Hence, allocations are important not only in dampening market power effects but play a dual role, also working to incentivize adequate investment from dealers into areas like facilities renovation or advertising.

As a whole, this evidence would suggest that sparse dealer networks, or equivalently concentration in dealer networks, is not a concern from an efficiency or welfare perspective. Classical results from the theory of vertical restraints show that in the case of a perfectly effective restraint, the market structure of downstream retailers is irrelevant. Results from the franchising literature suggest that some form of market power or protection is necessary to incentivize retailers to make appropriate non-price investments. An effective vertical restraint is able to balance these two opposing forces, mitigate pricing power from overly concentrated market structure, and minimize externality from an overly competitive market structure by shifting retailers’ private marginal costs of investment to be closer to the “social” marginal costs of the entire vertical structure. The practice of discretionary allocations appears to be a mechanism that is capable of striking this balance.

6 Conclusion

In this paper, I perform the first retrospective analysis of ownership changes among car dealerships. Despite these transactions leading to large changes in local market share, I find that no price or quantity changes result. I posit that the practice of allocations is likely the mechanism by which manufacturers are able to mediate market power effects, and I provide the first empirical evidence of which I am aware on the importance of trim allocation to dealer profitability. I also show evidence of positive spillovers between dealers of the same franchise, confirming the hypothesis of Murry 2018. I pair the empirical evidence from my analysis with a legal case study that makes explicit the role of allocations and how manufacturers use them to incentivize dealer behavior. Taken together, I claim that this should alleviate some of the concern regarding the strictness of state franchise laws and the increasing concentration among car dealerships.

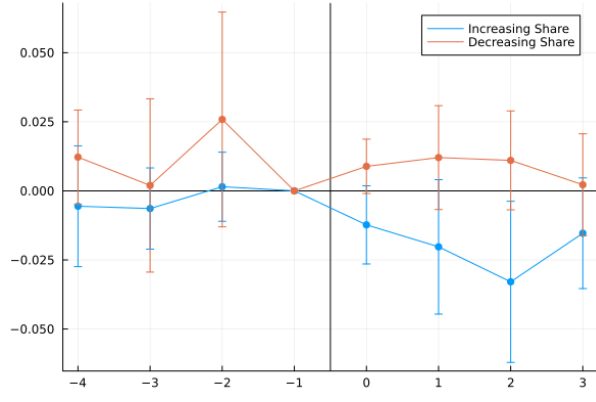
Despite the strength of this assertion, I maintain that a vertically integrated retail model is likely more efficient than the current status quo. Barring extreme circumstances of managerial ability or local expertise as mentioned in Lafontaine and Margaret E Slade 2015, sales via a vertically integrated distribution channel are almost certainly more efficient. However, the thrust of the evidence I present is to suggest manufacturers have in allocations a highly effective mechanism at their disposal that went previously unrecognized in the academic literature. Given manufacturers’ ability to discipline dealers via allocations—and assuming their

ongoing legal privilege to do so—it is likely that the efficiency loss from a fragmented distribution system is smaller than was previously thought.

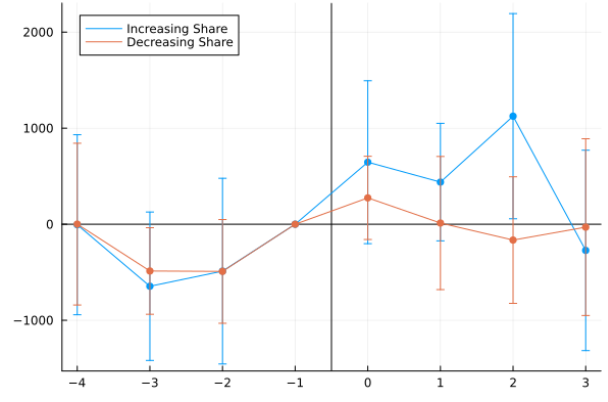
References

- Albuquerque, Paulo and Bart J. Bronnenberg (Jan. 2012). “Measuring the Impact of Negative Demand Shocks on Car Dealer Networks”. In: *Marketing Science* 31.1, pp. 4–23.
- Arkhangelsky, Dmitry et al. (Dec. 2021). “Synthetic Difference-in-Differences”. In: *American Economic Review* 111.12, pp. 4088–4118.
- Asker, John and Heski Bar-Isaac (Feb. 2014). “Raising Retailers’ Profits: On Vertical Practices and the Exclusion of Rivals”. In: *American Economic Review* 104.2, pp. 672–686.
- Brand, Keith, Chris Garmon, and Ted Rosenbaum (Nov. 2023). “In the Shadow of Antitrust Enforcement: Price Effects of Hospital Mergers from 2009 to 2016”. In: *The Journal of Law and Economics* 66.4, pp. 639–669.
- Bresnahan, Timothy F. and Peter C. Reiss (Oct. 1991). “Entry and Competition in Concentrated Markets”. In: *Journal of Political Economy* 99.5, pp. 977–1009.
- Brunson, Kendrick and Steve Barnes (2020). “Automotive Dealership Management Fundamentals”. In: *Liberty University Faculty Presentations and Publications*.
- Burgdorf, Jacob (2021). *Franchise Termination Laws, Craft Brewery Entry and Growth*. Working Paper EAG 21-3. Economic Analysis Group Discussion Paper.
- Chen, Chia-Wen and Shiou Shieh (2016). “Does Exclusive Dealing Matter? Evidence from Distribution Contract Changes in The U.S. Beer Industry”. In: *The Journal of Industrial Economics* 64.3, pp. 411–435.
- Holcomb, Richard (n.d.). *Formal Evidentiary Hearing: Tesla Motors, Inc., File No. 2016-001 - Request for a Hearing Pursuant to Va. Code §§ 46.2-1572(4) and 46.2-1573 to Be a Dealer*.
- Lafontaine, Francine and Fiona Scott Morton (Sept. 2010). “Markets: State Franchise Laws, Dealer Terminations, and the Auto Crisis”. In: *Journal of Economic Perspectives* 24.3, pp. 233–250.
- Lafontaine, Francine and Margaret E Slade (2015). “Franchising and Exclusive Distribution: Adaptation and Antitrust”. In: *The Oxford Handbook of International Antitrust Economics. Vol. 1*. Oxford Univ. Press.
- (May 1998). *Incentive Contracting and the Franchise Decision*. Working Paper. National Bureau of Economic Research: [6544](#).
- Lee, Robin S., Michael D. Whinston, and Ali Yurukoglu (Jan. 2021). “Chapter 9 - Structural Empirical Analysis of Contracting in Vertical Markets”. In: *Handbook of Industrial Organization*. Ed. by Kate Ho,

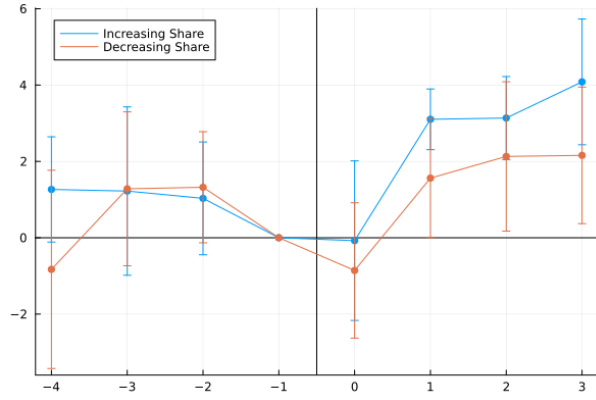
- Ali Hortaçsu, and Alessandro Lizzeri. Vol. 4. Handbook of Industrial Organization, Volume 4. Elsevier, pp. 673–742.
- Mankiw, N. Gregory and Michael D. Whinston (1986). “Free Entry and Social Inefficiency”. In: *The RAND Journal of Economics* 17.1, p. 48.
- Martín-Herrán, Guiomar, Simon Pierre Sigué, and Georges Zaccour (2014). “Downstream Horizontal Integration and Multiunit Dealership”. In: *International Transactions in Operational Research* 21.1, pp. 81–101.
- Mathewson, G Frank and Ralph A Winter (1984). “An Economic Theory of Vertical Restraints”. In: *The RAND Journal of Economics*, pp. 27–38.
- Murry, Charles (July 2018). “The Effect of Retail Competition on Relationship-Specific Investments: Evidence from New Car Advertising”. In: *International Journal of Industrial Organization* 59, pp. 253–281.
- Murry, Charles and Peter Newberry (Feb. 2022). “Franchise Contract Regulations and Local Market Structure”. In: *The Journal of Law and Economics* 65.1, pp. 105–130.
- Nurski, Laura and Frank Verboven (July 2016). “Exclusive Dealing as a Barrier to Entry? Evidence from Automobiles”. In: *The Review of Economic Studies* 83.3, pp. 1156–1188.
- Rey, Patrick and Joseph Stiglitz (1995). “The Role of Exclusive Territories in Producers’ Competition”. In: *The RAND Journal of Economics* 26.3, pp. 431–451. JSTOR: [2555997](#).
- Rey, Patrick and Jean Tirole (1986). “The Logic of Vertical Restraints”. In: *The American economic review*, pp. 921–939.
- Sass, Tim R. (Apr. 2005). “The Competitive Effects of Exclusive Dealing: Evidence from the U.S. Beer Industry”. In: *International Journal of Industrial Organization* 23.3, pp. 203–225.
- Scott Morton, Fiona and Ann McDermott (2017). “Tesla v. State Vehicle Franchise Laws”. In: *The Antitrust Revolution*. 6th Edition. Oxford Univ. Press.
- Smith, Richard L. (Apr. 1982). “Franchise Regulation: An Economic Analysis of State Restrictions on Automobile Distribution”. In: *The Journal of Law and Economics* 25.1, pp. 125–157.
- Sun, Liyang and Sarah Abraham (Dec. 2021). “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects”. en. In: *Journal of Econometrics*. Themed Issue: Treatment Effect 1 225.2, pp. 175–199.
- Tirole, Jean (1988). *The theory of industrial organization*. Cambridge, Mass: MIT Press.
- Verboven, Frank and Biliiana Yontcheva (Nov. 2024). “Private Monopoly and Restricted Entry—Evidence from the Notary Profession”. In: *Journal of Political Economy* 132.11, pp. 3658–3707.
- Zanarone, Giorgio (Nov. 2009). “Vertical Restraints and the Law: Evidence from Automobile Franchising”. In: *The Journal of Law and Economics* 52.4, pp. 691–700.



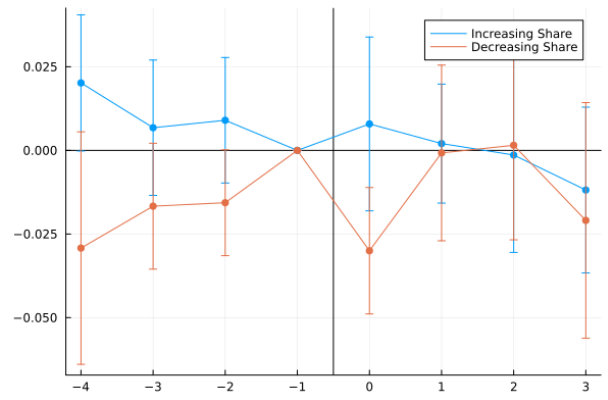
(a) Median Markup



(b) Mean Baseprice

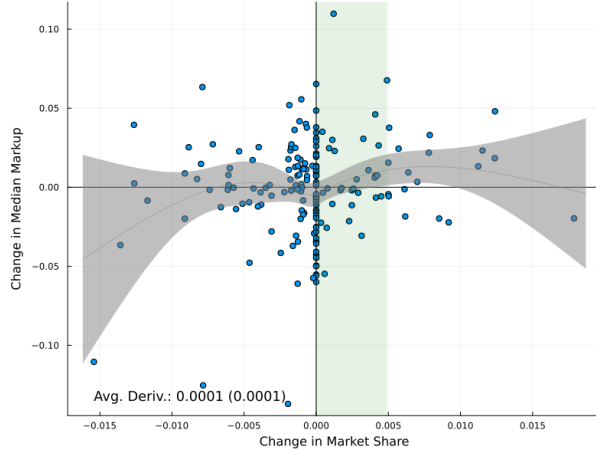


(c) Num. Unique Trims

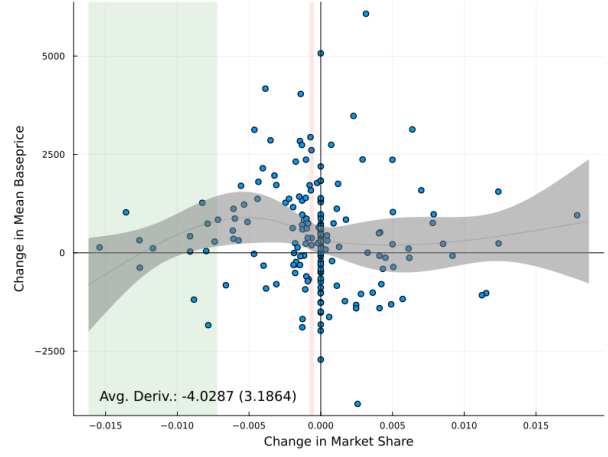


(d) Mean Trim Rank

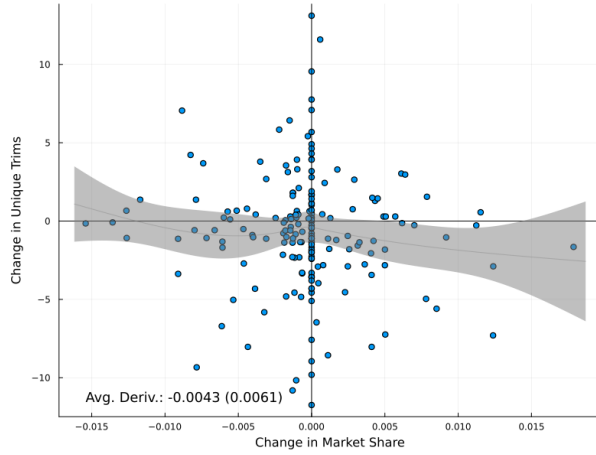
Figure 12: Event study figures for related outcomes of treated dealers, by whether the transaction increased or decreased local market share. Markups fell and number of unique trims increased for share-increasing transactions. All other diff-in-diff estimates are statistically insignificant.



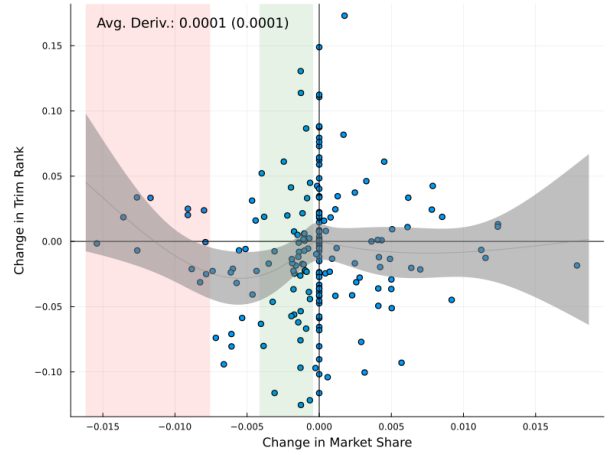
(a) Median Markup



(b) Mean Baseprice



(c) Number Unique Trims



(d) Mean Trim Rank

Figure 13: Difference-in-difference effects on other outcomes for affiliated dealerships, given the change in market share they experience because of a transaction.